

Toward Detecting Postpartum Depression Using Passive Mobile Sensing: Exploratory Analysis

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Abstract

Postpartum depression (PPD) is a prevalent mental health issue with severe consequences, yet its early detection is hampered by limitations in traditional screening methods like self-reported questionnaires. This study explores the potential of passive mobile sensing (PMS) to objectively assess PPD symptoms in postpartum women in Japan. Our iOS application, ZeroPPD, continuously collected sensor-derived behavioral metrics (e.g., mobility, activity, and phone usage) from 21 participants over four weeks. During this period, participants also submitted self-reported data on infant and maternal sleep, along with periodic Edinburgh Postnatal Depression Scale (EPDS) scores. Statistical analyses, including Linear Mixed-effects Models, identified significant correlations between specific behavioral features and EPDS scores. Key findings revealed specific behavioral patterns associated with PPD, including infant-related sleep disruptions and maternal caregiving stress (e.g., increased morning baby crying, longer nighttime stay durations), altered daytime activity and stationary behaviors (e.g., frequent afternoon activity bouts linked to lower EPDS), and potentially stress-related mobility. These results highlight PPD-specific digital biomarkers, suggesting that PMS can offer an effective pathway for continuous, objective monitoring of PPD-related behavioral changes, paving the way for the subsequent development of data-driven early risk identification tools and timely interventions.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Smartphones**; **Mobile phones**; **Mobile devices**.

Keywords

Passive Mobile Sensing; Postpartum Depression; Behavioral Data; Digital Biomarkers; Exploratory Analysis

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1 Introduction

Postpartum depression (PPD) is a prevalent and serious mental health condition affecting approximately 10–15% of mothers within three months postpartum [14]. It can manifest through symptoms such as sleep disturbances, irritability, and anhedonia, potentially leading to severe outcomes like self-harm or suicidal ideation [3, 19]. The postpartum period introduces unique psychological challenges due to hormonal shifts, increased caregiving demands, and altered social dynamics, heightening the risk of mental health disorders [21, 30]. In Japan, mental health issues have alarmingly become a leading cause of maternal mortality since 2020 [8], and globally, women with perinatal depression face a significantly higher mortality risk [11]. PPD's impact extends beyond the mother, affecting spousal well-being, mother-infant bonding, and child development [8].

Despite its severe consequences, early PPD detection remains challenging. Current screening predominantly relies on subjective self-reported questionnaires like the Edinburgh Postnatal Depression Scale (EPDS) [7]. However, these methods face several significant limitations, prompting the need for innovative approaches.

To address these limitations, passive mobile sensing (PMS) offers a promising avenue for continuous and objective mental health monitoring by analyzing behavioral patterns derived from smartphone data. This study explores the potential of PMS for assessing postpartum mental health conditions, aiming to identify behavioral indicators correlated with PPD symptoms and investigate how these can be used to develop effective assessment tools for this vulnerable population.

2 Background and Objectives

2.1 Challenges in PPD Screening

Traditional PPD screening, primarily using self-reported questionnaires such as the EPDS [7], is a cornerstone of current practice. While the EPDS is widely adopted, its reliance on self-assessment

presents several inherent limitations. Individuals may struggle to accurately recall symptoms (recall bias) or alter responses due to privacy concerns or stigma (social desirability bias) [20]. Furthermore, the periodic administration of these questionnaires makes it difficult to capture the dynamic nature of mental health states or provide continuous monitoring, potentially delaying timely intervention [1, 23].

Beyond these issues, traditional screening methods often face challenges in widespread implementation. These can include significant time and financial costs, limited public awareness of PPD and screening options, and difficulties in accessing specialized services [7]. For new mothers already navigating demanding schedules and significant life adjustments, the logistical hurdles of seeking formal assessments can be particularly prohibitive, leading to underdiagnosis and untreated PPD. These multifaceted limitations underscore the urgent need for more accessible, objective, and continuous PPD screening approaches.

2.2 PMS for PPD Assessment: Our Approach

PMS has emerged as a powerful alternative for mental health assessment, leveraging digital biomarkers—quantifiable physiological and behavioral measures passively collected from digital devices—to infer mental well-being [5, 13, 25]. This approach allows for dynamic, objective, and continuous monitoring [10, 12]. Research in general mental health has linked various digital biomarkers to depressive symptoms, including mobility patterns (e.g., reduced activity space [22, 24, 29]), daily activity and sedentary behavior (e.g., irregular routines, increased sedentary time [2, 4, 26]), sleep patterns (e.g., variations in duration, irregular schedules [1, 4, 27]), social interaction (e.g., reduced call frequency/duration [17, 28]), and overall device usage [17, 29].

However, applying this approach to Postpartum Depression (PPD) presents unique challenges. While initial studies confirm the feasibility of PMS in postpartum populations [9, 15], they have yet to identify statistically significant PPD-specific biomarkers. Critically, biomarkers from general depression are not directly transferable. For example, increased sedentary time may reflect essential infant care like breastfeeding rather than apathy, and fragmented sleep is often a normal result of attending to a newborn, not simply a sign of a sleep disorder. Therefore, a critical gap exists in identifying and interpreting digital biomarkers specifically for PPD.

This research aims to address this gap. Our study focuses on the following research questions:

- RQ1: Which PMS-derived digital biomarkers are significantly correlated with PPD symptoms, as measured by the EPDS?
- RQ2: How do these biomarkers reveal PPD-specific risk patterns that reflect the unique context of the postpartum period?

To answer these questions, our study makes the following contributions. Through the collection and analysis of behavioral data from postpartum women, we provide:

- The construction of a novel dataset capturing the behavioral patterns of postpartum women via PMS.
- The identification and interpretation of key PPD-specific digital biomarkers, distinguishing them from markers of general depression.

This work lays a foundation for data-driven PPD monitoring and tailored mental health screening for the postpartum population.

3 Methods

3.1 Participants and Data Collection

Postpartum women in Japan were recruited via a crowdsourcing service in Japan (i.e., Yahoo! Crowdsourcing [6]) for this study. Participants were selected through a two-stage screening process based on the following criteria: age twenty years or older, raising an infant less than one year old, and use of a compatible iOS device. This study was reviewed and approved by the Institutional Review Board of the Center for Spatial Information Science, The University of Tokyo.

We developed ZeroPPD, an iOS application based on the AWARE framework [18], for data collection. ZeroPPD gathered behavioral data (steps, accelerometer, GPS, screen state, calls, activity recognition) from smartphone sensors and self-reported data on infant and maternal sleep, and periodically administered the Japanese version of the EPDS [16]. Demographic and contextual data were collected during the recruitment process. All collected data were anonymized and securely uploaded to our SQL database on a cloud service. The formal data collection spanned 4 weeks (December 2nd - 29th, 2024), preceded by a brief setup period.

3.2 Feature Engineering

Behavioral features were primarily extracted on a weekly basis, with data segmented by workday/holiday status and distinct times of day (morning, afternoon, evening, and night). For sensor readings within these segments, standard aggregate statistics (mean, standard deviation, max, and median) were computed.

Beyond general aggregations, specific data types underwent tailored processing, with feature engineering methodologies informed by relevant research [4, 29]. For instance, raw location data were preprocessed to identify stay-points, which were then clustered using a density-based algorithm. This facilitated the derivation of spatial behavior features, such as the number of significant locations, time spent at these locations, location entropy, and mobility patterns like the radius of gyration. The Lomb-Scargle periodogram was used on location-derived metrics to assess temporal regularity in movement, such as circadian rhythms. Screen interaction patterns were quantified by extracting features like unlock frequency, unlocked session duration, and first/last interaction times. Similarly, activity recognition data, including types like walking, running, automotive, and cycling, were processed. This involved categorizing continuous periods as either stationary bouts (stationary=1) or active bouts (stationary=0, reflecting engagement in the specified activities), yielding metrics on their count and duration. Daily morning self-reports of baby crying and maternal awakenings were aggregated weekly, applying temporal segmentation and statistical summarization comparable to that used for sensor-derived features.

3.3 Feature Selection and Statistical Analysis

To identify behavioral features associated with EPDS scores, we first performed dimensionality reduction using low-variance filtering and regularization-based feature selection with internal cross-validation. Subsequently, for each selected feature, we constructed

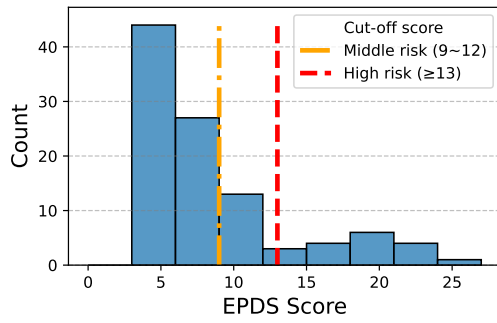


Figure 1: Weekly EPDS scores among participants (N=21).

a Linear Mixed-effects Model (LMM) to predict EPDS scores. These models, confirmed to meet all key statistical assumptions, included the behavioral feature as a fixed effect, participant ID as a random intercept, and controlled for covariates such as family income and maternal age. To account for multiple comparisons inherent in analyzing numerous features from these LMMs, the Benjamini-Hochberg False Discovery Rate (FDR) correction was applied to the p-values, with features having FDR-corrected $q < 0.05$ and $q < 0.1$ considered significant.

4 Results

4.1 Evaluation Procedure

This study was initiated with over 700 raw features, which were reduced to approximately 70, leading to univariable LMM analyses to assess their association with weekly EPDS scores. The study included 21 participants, and their weekly EPDS scores (Figure 1) revealed a mean score of 7.81 (SD=5.34), with 23.81% of participants scoring above the high-risk threshold of 13, indicating a notable level of depressive symptoms within the sample.

4.2 Significant Features

After FDR correction, we identified features significantly associated with EPDS scores. A total of 8 features were significant at $q < 0.05$, with an additional 2 features showing a potential trend ($0.05 \leq q < 0.10$). The top significant features, ranked by q-value and displaying effect size (standardized regression coefficient β), are presented in Figure 2, with their full names provided below.

Key findings include:

Higher baby crying counts on weekdays (*baby-crying-weekday-median*; $\beta = +0.2092$, $q < 0.05$), based on users' daily reports, were associated with higher EPDS scores, suggesting a direct negative impact of infant sleep issues on maternal mood. Some PMS features also showed significant correlations: Longer stay duration at a single location during the night (00:00-06:00) (*gps-max-stay-duration-night-whole-week*; $\beta = +0.2962$, $q < 0.05$) also correlated with increased EPDS scores; this particular finding, while statistically significant, warrants cautious interpretation. Conversely, more total active bouts in weekday afternoons (*activity-total-active-bouts-mean-afternoon-weekday*; $\beta = -0.2436$, $q < 0.05$) were linked to lower EPDS scores, indicating that multiple short bursts of physical

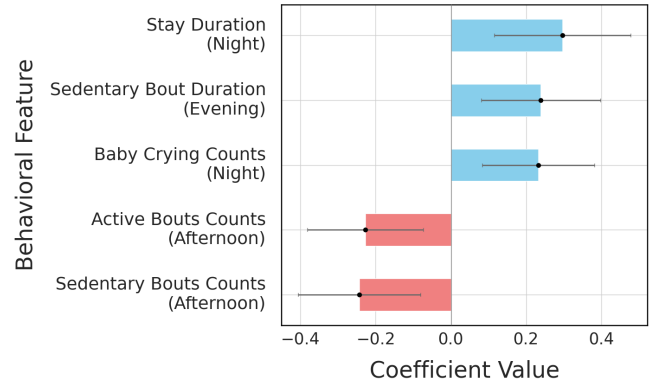


Figure 2: Top features significantly associated with weekly EPDS scores (FDR-corrected). Bar length indicates the absolute standardized regression coefficient β (effect size); color/direction indicates positive or negative association. Feature names are simplified (full names in text). Error bars depict 95% Confidence Intervals.

activity in the afternoon may be beneficial for mitigating depressive symptoms.

Further significant features included the duration of stationary bouts (*activity-avg-stationary-bout-duration-mean-night-weekday*; $\beta = +0.1307$, $q < 0.05$), and the number of stationary bouts (*activity-total-stationary-bouts-mean-afternoon-weekday*; $\beta = -0.2277$, $q < 0.05$).

A detailed interpretation of these associations will be provided in the Discussion section.

5 Discussion

Our findings suggest three core behavioral patterns associated with perinatal depression, which provide nuanced insights into the daily life manifestations of perinatal mental health:

Sleep Deprivation and Caregiving Stress: Increased nighttime baby crying and frequent awakenings indicate poor maternal sleep and caregiving stress, and we explored associated sensor-derived behaviors. Notably, longer maximum nighttime stays at one location positively correlated with EPDS scores. Accelerometer data crucially reinforced this, showing longer average weekday night stationary bouts also correlated with higher EPDS scores. Contrary to the intuition that longer nighttime stillness at home signifies better sleep, our findings link this indirectly-inferred prolonged stillness to higher EPDS scores. Although lacking direct sleep measures, combined sensor data suggest this stillness may not be restorative. Instead, this positive correlation might reflect issues such as extended, low-quality nighttime infant care; maternal sleep difficulties due to anxiety or rumination; or broader depressive behavioral restriction. Therefore, despite the limitations of indirect sleep assessment, the consistent association with higher EPDS scores supports problematic nocturnal experiences rather than restful sleep.

Daytime Activity and Stationary Patterns: Lower EPDS scores were associated with more total active bouts in weekday

afternoons (negative correlation) and, interestingly, more total stationary bouts in weekday afternoons (negative correlation), suggesting that frequent interruptions of stationary time, rather than total stationary time alone, might be protective. Conversely, longer average duration of stationary bouts, especially in the evening on holidays, was linked to higher depression scores. This highlights the importance of the *pattern* of activity and sedentariness during daytime and evening hours.

Potential Stressful Mobility Pattern: An exploratory finding indicates that greater median travel distance on weekday mornings was potentially associated with increased depressive symptoms ($0.05 \leq q < 0.10$). This contrasts with typical findings of reduced mobility in depression and may suggest that certain trips during the perinatal period are stress-inducing rather than restorative.

5.1 Comparison with Existing Literature

Our results align with previous research showing that reduced physical activity and increased stationary behavior are linked to depression [2, 4, 22, 26], supported by our findings on afternoon activity and evening stationary duration. Similarly, sleep problems [1, 4, 27], evidenced by maternal night awakenings, are confirmed risk factors.

This study also offers unique insights specific to the perinatal period, moving beyond foundational feasibility explorations in postpartum populations [15] to identify distinct PPD-related behavioral patterns and digital biomarkers. The strong predictive power of infant-related sleep disruptions (e.g., baby crying counts) for maternal EPDS scores is a key perinatal-specific finding. The interpretation of prolonged nighttime stillness—supported by both GPS-defined long stays and accelerometer-defined long stationary bouts—is complex. As discussed, rather than simply indicating rest, it may reflect nuanced challenges of the perinatal period, such as demanding infant care schedules or maternal sleep disturbances. This highlights that sensor-derived nighttime stillness requires cautious interpretation in the absence of direct sleep measures, though the convergent evidence from two distinct sensor types strengthens the hypothesis of it being a marker of distress. The potential link between increased morning travel distance and higher EPDS scores also challenges the common "activity space shrinkage" narrative in depression, hinting at possible stress associated with perinatal-specific travel. Furthermore, while increased stationary time is a known risk, our finding that a higher *number* of afternoon stationary bouts (implying more frequent breaks in stationary time) is associated with lower EPDS scores, alongside longer *average duration* of stationary bouts being a risk, underscores the nuanced role of stationary patterns, which warrants further investigation.

The non-significance of some expected factors, such as direct social interaction metrics (e.g., call frequency/duration) [17, 28], overall mobile phone usage [17, 29], or total daily step counts [22, 24, 29], might be due to the unique life circumstances of perinatal women, where infant care overwhelmingly shapes routines, or due to limitations in our data's granularity (e.g., lack of context for social interactions).

5.2 Limitations and Future Directions

For perinatal women, these findings underscore the importance of addressing infant sleep, protecting maternal sleep quality (the nature of which requires more than just duration at a location or general inactivity), engaging in frequent short bouts of afternoon activity, and breaking up long periods of stationary behavior. Clinically, these behavioral digital phenotypes offer potential targets for early identification and intervention, though features reflecting nighttime stillness require careful contextual interpretation.

Limitations: While this study offers initial insights into digital phenotypes associated with perinatal mental health, several methodological considerations warrant discussion. This study's findings should be considered in light of its potentially modest sample size ($N=21$), affecting statistical power, and its exploratory nature. The stringent FDR correction might have led to some false negatives. Data limitations include the absence of direct objective sleep measures (e.g., total sleep time, sleep efficiency), which, as highlighted, makes the interpretation of features like GPS-defined nighttime stays and accelerometer-derived nighttime stationary duration particularly challenging in terms of definitively assessing sleep quality, though their combined association with higher EPDS scores provides a stronger signal of problematic nocturnal experiences. Other limitations include a lack of detailed contextual information on social interactions and the indirect inference of some behaviors from sensor data. Collectively, these factors highlight the exploratory nature of our conclusions and emphasize the need for caution when generalizing these results.

Future Directions: Future research should validate these findings in larger, more diverse samples while integrating multi-modal data (e.g., clinical interviews, direct sleep monitoring) to provide deeper, contextualized insights into sensor-derived behaviors and their underlying mechanisms. We also recommend a more nuanced investigation of "stressful mobility" and nighttime behaviors using combined sensor data. As an intermediate outcome, the key behavioral features from this study lay the groundwork for developing comprehensive models to predict PPD risk and aid early screening. Ultimately, these digital phenotypes and models could inform personalized intervention strategies for perinatal mental health.

6 Conclusion

This study explored PMS via smartphones to objectively assess PPD symptoms in Japanese mothers. Using the ZeroPPD app, we collected behavioral data and EPDS scores, identifying significant digital biomarkers. Key findings revealed PPD-associated patterns such as infant-related sleep disruptions, caregiving stress, altered daytime activity, stationary behaviors, and potential stress-linked mobility. These results highlight PMS's utility for continuous, objective monitoring of PPD-related behavioral changes, offering advantages over traditional screening. The identified biomarkers are crucial for developing data-driven tools for early PPD risk identification and timely interventions, aiming to improve maternal mental health.

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